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INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Programme: **B.E / B.Tech**

Course Code: **MLA02**

Course Name: **FUNDAMENTALS OF MACHINE LEARNING**

Duration: **1:30 Hr**

Date : _____ Max Marks: **50 Marks**

ASSIGNMENT – IV

UNIT IV – PROBABILISTIC GRAPHICAL MODELS

Topics: *Directed and undirected graphical models, Bayesian networks, conditional independence, Markov random fields, inference, learning and generalisation, hidden Markov models, conditional random fields*

Course Outcome

On successful completion of this assignment, students will be able to:

CO	Course Outcome Statement
CO4	Develop probabilistic graphical models for performing inference, learning patterns, and analyzing sequential data.

Answer all the Tasks

Task 1. A home alarm network has binary variables Burglary (B), Earthquake (E) and Alarm (A) with edges $B \rightarrow A$ and $E \rightarrow A$. $P(B) = 0.001$, $P(E) = 0.002$ and $P(A = 1 \mid B, E) = 0.95$ (B=1,E=1), 0.94 (B=1,E=0), 0.29 (B=0,E=1), 0.001 (B=0,E=0).

(a) Write the factorised joint distribution. (b) Compute $P(A = 1)$. (c) Compute $P(B = 1 \mid A = 1)$. (d) Compute $P(B = 1 \mid A = 1, E = 1)$ and explain the 'explaining-away' effect.

(5) (CO4) (BL4)

Task 2. Five binary variables A, B, C, D, E form the DAG $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$, $C \rightarrow E$. Calculate the number of independent parameters needed by this Bayesian network and compare it with an unrestricted joint distribution over the five variables. State which conditional independence assumption produces the saving.

(2) (CO4) (BL4)

Task 3. Consider the directed acyclic graph with edges $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$, $C \rightarrow E$, $D \rightarrow F$ and $E \rightarrow F$. Using d-separation, state with justification whether each of the following statements is true (✓) or false (✗):

(a) $A \perp B \mid \varnothing$

(b) $A \perp B \mid \{C\}$

(c) $A \perp B \mid \{F\}$

(d) $D \perp E \mid \{C\}$

(e) $D \perp E \mid \{C, F\}$

(f) $A \perp F \mid \{C\}$

(6) (CO4) (BL4)

Task 4. A pairwise Markov random field over binary variables $X_1 — X_2 — X_3$ (a chain) has potentials $\psi(x_i, x_{i+1}) = 3$ if $x_i = x_{i+1}$ and 1 otherwise, for both edges. (a) Compute the partition function Z by enumeration. (b) Compute $P(X_2 = 1)$ and $P(X_1 = X_3)$. (c) Compute $P(X_1 = 1 \mid X_3 = 1)$ and explain how conditioning on X_2 changes the dependence between X_1 and X_3.

(4) (CO4) (BL4)

Task 5. In the chain $A \rightarrow B \rightarrow C$ of binary variables, $P(A=1) = 0.4$; $P(B=1|A=0) = 0.3$, $P(B=1|A=1) = 0.8$; $P(C=1|B=0) = 0.1$, $P(C=1|B=1) = 0.7$. Use variable elimination to compute (a) the message $\tau(B) = \Sigma_a P(a)P(B|a)$, (b) $P(C = 1)$, and (c) $P(A = 1 \mid C = 1)$. Explain the computational advantage of elimination over enumerating the joint.

(3) (CO4) (BL4)

Task 6. From 20 records of the network Rain (R) → Sprinkler (S), the following counts were observed: among the 8 days with R = 1 the sprinkler was on (S = 1) on 1 day; among the 12 days with R = 0 it was on on 6 days. (a) Compute the maximum-likelihood estimates of $P(S=1|R=1)$ and $P(S=1|R=0)$. (b) Recompute with Laplace (add-one) smoothing. (c) Compute the log-likelihood of the 12 + 8 sprinkler outcomes under the ML parameters, and explain why smoothing is preferred for small counts.

(3) (CO4) (BL4)

Common data for Tasks 7, 8 and 9 – a two-state weather HMM. Hidden states: Rainy (R), Sunny (S). Observations: Walk, Shop, Clean. Observed sequence O = (Walk, Shop, Clean).

Initial π	Rainy	Sunny
$P(q_1)$	0.6	0.4

Transition A	→ Rainy	→ Sunny
From Rainy	0.7	0.3
From Sunny	0.4	0.6

Emission B	Walk	Shop	Clean
Rainy	0.1	0.4	0.5
Sunny	0.6	0.3	0.1

Task 7. Using the HMM defined above, apply the forward algorithm to compute $\alpha_1(i)$, $\alpha_2(i)$ and $\alpha_3(i)$ for both states and hence the likelihood $P(O \mid \lambda)$ of the sequence (Walk, Shop, Clean). Explain the complexity gain over summing over all state sequences.

(4) (CO4) (BL4)

Task 8. For the same HMM and observation sequence, use the Viterbi algorithm to find the most probable hidden state sequence. Show $\delta_3(i)$ and the back-pointers, report the probability of the best path, and explain how Viterbi differs from the sequence of individually most likely states.

(4) (CO4) (BL4)

Task 9. For the same HMM and observation sequence, compute the backward variables $\beta_3(i)$ and $\beta_2(i)$ and use the forward–backward result to find the posterior probabilities $\gamma_2(i) = P(q_2 = i \mid O, \lambda)$ of the weather on day 2.

(3) (CO4) (BL4)

Task 10. A linear-chain conditional random field labels a two-token sentence with tags in {B, I}. The unary (emission) scores are $s_1(B) = 2.0$, $s_1(I) = 0.5$ for token 1 and $s_2(B) = 0.5$, $s_2(I) = 1.5$ for token 2. The transition scores are $w(B \to B) = 0.2$, $w(B \to I) = 1.0$, $w(I \to B) = 0.3$, $w(I \to I) = 0.8$. The unnormalised log-score of a tag sequence is the sum of its unary and transition scores. (a) Compute the score of all four sequences. (b) Compute the partition function $Z(x)$ and the probability of each sequence. (c) Identify the best sequence and explain why a CRF avoids the label-bias and independence assumptions of an HMM.

(4) (CO4) (BL4)

Task 11. Two candidate Bayesian-network structures were learned from $N = 200$ records. Structure M_1 has $k = 7$ free parameters and maximised log-likelihood -420; M_2 has $k = 15$ parameters and log-likelihood -410. Compute the penalised scores $BIC = \ln L - (k/2)\ln N$ and $AIC = \ln L - k$ (higher is better) for both, evaluate which structure each criterion selects, and justify a final choice with reference to generalisation.

(4) (CO4) (BL5)

Task 12. A clinic models lung-cancer risk with binary variables Pollution (P = high), Smoker (S), Cancer (C), X-ray result (X = positive) and Dyspnoea (D). Domain experts propose that Pollution and Smoker influence Cancer, and Cancer influences both X-ray and Dyspnoea.

Parameters: $P(P=1) = 0.1$; $P(S=1) = 0.3$; $P(C=1 \mid P, S) = 0.05$ $(P=1,S=1)$, 0.02 $(P=1,S=0)$, 0.03 $(P=0,S=1)$, 0.001 $(P=0,S=0)$; $P(X=1 \mid C=1) = 0.9$, $P(X=1 \mid C=0) = 0.2$; $P(D=1 \mid C=1) = 0.65$, $P(D=1 \mid C=0) = 0.30$.

(a) Draw the network structure (list its edges) and write the factorised joint distribution. [1]

(b) Compute the prior probability $P(C = 1)$ by marginalising out P and S. [2]

(c) A positive X-ray is observed. Compute $P(C = 1 \mid X = 1)$. [2]

(d) The patient also reports dyspnoea. Compute $P(C = 1 \mid X = 1, D = 1)$. [2]

(e) Using the graph, justify why X and D can be treated as conditionally independent given C, and evaluate how the diagnosis would change if the patient were also known to be a smoker (compute $P(C = 1 \mid X = 1, S = 1)$). [1]

(8) (CO4) (BL6)

Task 1:

given,

$P(B) = 0.001$, $P(E) = 0.002$

$P(A=1 \mid B, E) = 0.95$

$P(A=1 \mid B=1, E=0) = 0.94$

$P(A=1 \mid B=0, E=1) = 0.29$

$P(A=1 \mid B=0, E=0) = 0.001$

(a)

The joint probability distribution is:

$$P(B, E, A) = P(B)\, P(E)\, P(A \mid B, E)$$

(b)

$$P(A) = \sum_{B,E} P(B)\, P(E)\, P(A \mid B, E)$$

$$= (0.001)(0.002)(0.95) + (0.001)(0.998)(0.94) + (0.999)(0.002)(0.29) + (0.999)(0.998)(0.001)$$

$$= 0.00000019 + 0.00093812 + 0.00057942 + 0.000997002$$

$$= 0.00251644 2$$

$$P(A=1) = 0.00251644 2$$

(c)

using Baye's theorem

$$P(B \mid A) = \frac{P(B, A)}{P(A)}$$

$$P(B=1, A=1)$$

$$= P(B=1)\, P(E=1)\, P(A \mid B=1, E=1) + P(B=1)\, P(E=0) P(A \mid B=1, E=0)$$

$= (0.001)(0.002)(0.95) + (0.001)(0.998)(0.94)$

$= 0.00094002$

$$P(B=1 \mid A=1) = \frac{0.00094002}{0.00251642}$$

$$\boxed{P(B=1 \mid A=1) = 0.37355}$$

(d)

Using Baye's Theorem:

$$P(B \mid A, E) = \frac{P(B)\,P(A \mid B, E)}{P(A \mid E)}$$

$P(A=1 \mid E=1)$

$= (0.001)(0.95) + (0.999)(0.29)$

$= 0.00095 + 0.28971$

$= 0.29066$

$P(B=1 \mid A=1, E=1)$

$$= \frac{0.001 \times 0.95}{0.29066}$$

$$\boxed{P(B=1 \mid A=1, E=1) = 0.003268}$$

Explaining-away effect:

When the alarm rings, both burglary and earthquake are possible causes. If we already know that an earthquake occurred the earthquake explains why the alarm rang. Therefore, the probability of burglary becomes much smaller.

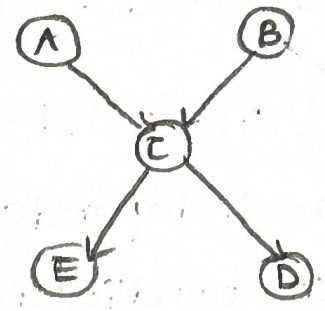
This is called the explaining-away effect.

Task 2:

Given,

Five Binary variables:

A, B, C, D, E



Network structure:

$A \rightarrow C$, $B \rightarrow C$

$C \rightarrow D$, $C \rightarrow E$

For a binary variable:

$$\text{Parameters} = 2^{\text{number of parents}}$$

variable	Parents	Parameters
A	0	1
B	0	1
C	2	4
D	1	2
E	1	2

Total Parameters = 1+1+4+2+2

= 10

Unrestricted Joint distribution

For 5 binary variables

$$\text{Parameters} = 2^5 - 1$$

$$= 32 - 1$$

$$= 31$$

Parameters = 31

comparison

Bayesian network = 10 parameters

unrestricted joint distribution = 31 parameters

Parameters saved = 31 − 10 = 21

conditional independence assumptions:

A and B are independent

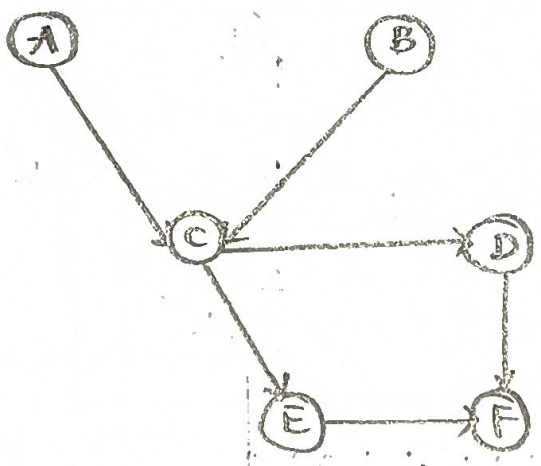
D and E are conditionally independent given C.

D and E depend only on C.

Task-3:

Given,

$A \rightarrow C, B \rightarrow C, C \rightarrow D, C \rightarrow E, D \rightarrow F, E \rightarrow F$



(a) $A \perp B \mid \emptyset$

True (✓)

Because,

→ A and B meets at C, which is a collider

→ A collider blocks the path when it is not observed.

Therefore, A and B are independent.

(b) $A \perp B \mid \{C\}$

False (X)

Because,

C is a collider

when C is observed, the path between A and B becomes active

Therefore, A and B become dependent.

(c) $A \not\perp B \mid \{F\}$

False (X)

F is a descendant of the collider C

Observing a descendant of a collider opens the path b/w A and B

Therefore, A and B are independent.

(d) $D \perp E \mid \{C\}$

True (✓)

The path between D and E through C is

$$D \leftarrow C \rightarrow E$$

C is a common parent

when C is observed, the path is blocked

The other path is:

$$D \rightarrow F \leftarrow E$$

F is a collider, which is not observed

Therefore, D and E are independent.

(e) $D \not\perp E \mid \{C, F\}$

False (X)

F is a collider between D and E

when F is observed, the path becomes active

Therefore, D and E are dependent

(f) $A \perp F \mid \{C\}$

True (✓)

All paths from A to F pass through C

when C is observed, these paths are blocked

Therefore, A and F are conditionally independent.

Task-4:

Given,

A chain of three binary variables:

$$x_1 - x_2 - x_3$$

Potential Function:

$$\psi(x_i, x_{i+1}) = \begin{cases} 3 & x_i = x_{i+1} \\ 1 & x_i \neq x_{i+1} \end{cases}$$

(a) compute the partition function Z:

The partition function is:

$$Z = \sum_{x_1, x_2, x_3} \psi(x_1, x_2)\, \psi(x_2, x_3)$$

$2^3 = 8$ possible combinations

x_1	x_2	x_3	weight
0	0	0	$3 \times 3 = 9$
0	0	1	$3 \times 1 = 3$
0	1	0	$1 \times 1 = 1$
0	1	1	$1 \times 3 = 3$
1	0	0	$1 \times 3 = 3$
1	0	1	$1 \times 1 = 1$
1	1	0	$3 \times 1 = 3$
1	1	1	$3 \times 3 = 9$

Adding all weights:

$$z = 9+3+1+3+3+1+3+9$$

$$= 32$$

$$\boxed{z = 32}$$

(b) compute $P(x_2=1)$ and $P(x_1=x_3)$

1. $P(x_2=1)$

Adding all weights where $x_2=1$:

$$= 1+3+3+9 = 16$$

$$P(x_2=1) = \frac{16}{32}$$

$$\boxed{P(x_2=1) = 0.5}$$

2. $P(x_1=x_3)$

Adding the weights where x_1 and x_3 are equal:

$$= 9+1+1+9$$

$$= 20$$

$$P(x_1=x_3) = \frac{20}{32}$$

$$\boxed{P(x_1=x_3) = 0.625}$$

(c) compute $P(x_1=1 \mid x_3=1)$

using conditional probability.

$$P(x_1=1 \mid x_3=1) = \frac{P(x_1=1, x_3=1)}{P(x_3=1)}$$

Numerator:

$$= 1+9 = 10$$

Denominator:

$$= 3+3+1+9 = 16$$

$$\therefore = \frac{10}{16}$$

$$P(x_1=1 \mid x_3=1) = 0.625$$

Explanation:

x_1 and x_3 are dependent when x_2 is not observed

However, when x_2 is given, x_1 and x_3 become conditionally independent.

This is because x_2 separates the two variables in the Markov network.

Task-5:

Given,

$$A \rightarrow B \rightarrow C$$

$P(A=1) = 0.4$

$P(B=1 \mid A=0) = 0.3$

$P(B=1 \mid A=1) = 0.8$

$P(C=1 \mid B=0) = 0.1$

$P(C=1 \mid B=1) = 0.7$

(a) calculate the message $T(B)$.

$$T(B) = \sum_A P(A)\, P(B \mid A)$$

For $B=1$:

$$P(B=1) = P(A=0)\, P(B=1 \mid A=0) + P(A=1)\, P(B=1 \mid A=1)$$

$$= (0.6)(0.3) + (0.4)(0.8)$$

$$= 0.18 + 0.32$$

$$P(B=1) = 0.5$$

For $B=0$:

$$P(B=0) = 1 - 0.5$$

$$P(B=0) = 0.5$$

$$\therefore T(B) = [0.5, 0.5]$$

(b) calculate $P(C=1)$

using total probability:

$$P(C=1) = \sum_B P(B)\, P(C|B)$$

$$= (0.5)(0.1) + (0.5)(0.7)$$

$$= 0.05 + 0.35$$

$$P(C=1) = 0.4$$

(c) calculate $P(A=1|C=1)$

using Bayes' theorem

$$P(A=1|C=1) = \frac{P(A=1, C=1)}{P(C=1)}$$

$$P(A=1, C=1)$$

$$= P(A=1) \sum_B P(B|A=1)\, P(C=1|B)$$

$$= 0.4\,[(0.2)(0.1) + (0.8)(0.7)]$$

$$= 0.4\,[0.02 + 0.56]$$

$$= 0.232$$

$$P(A=1|C=1) = \frac{0.232}{0.4}$$

$$\boxed{P(A=1|C=1) = 0.58}$$

computational advantage:

variable elimination reduces repeated calculations by eliminating intermediate variables one at a time. It is more efficient than calculating every possible combination in the joint distribution, especially for large networks.

Task-6:

Given,

Total records = 20

Rain(R)	Total Days	Sprinkler ON
R=1	8	1
R=0	12	6

(a) Maximum Likelihood Estimates

$$P(S=1|R) = \frac{\text{Number of sprinkler ON days}}{\text{Total days}}$$

For R=1:

$$P(S=1|R=1) = \frac{1}{8}$$

$$= 0.125$$

For R=0:

$$P(S=1|R=0) = \frac{6}{12}$$

$$= 0.5$$

(6) using Laplace smoothing

$$P(S=1|R) = \frac{\text{count}(S=1)+1}{\text{Total count}+2}$$

FOR R=1 :

$$= \frac{1+1}{8+2}$$

$$= \frac{2}{10}$$

$$= 0.2$$

FOR R=0 :

$$= \frac{6+1}{12+2}$$

$$= \frac{7}{14}$$

$$= 0.5$$

(c) log-likelihood

$$LL = \sum n_i \log(P_i)$$

$$LL = 1 \log(0.125) + 7 \log(0.875) + 6 \log(0.5) + 6 \log(0.5)$$

$$= -2.0794 - 0.9353 - 4.1589 - 4.1589$$

$$\boxed{LL = -11.3319}$$

Why is smoothing preferred?

→ It avoids zero probabilities

→ It reduces overfitting

→ It provides more reliable estimates for small datasets

→ It prevents unseen events from receiving zero probability.

7,8,9

Given,

Hidden states : Rainy(R), sunny(S)

observations : walk, shop, clean

observation sequence :

$$O = (walk, shop, clean)$$

Initial probabilities :

State	Probability
Rainy	0.6
sunny	0.4

Transition matrix :

From/To	Rainy	sunny
Rainy	0.7	0.3
sunny	0.4	0.6

Emission matrix :

state	walk	shop	clean
Rainy	0.1	0.4	0.5
sunny	0.6	0.3	0.1

Task-7:

$$\alpha_1(i) = \pi_i b_i(O_1)$$

$$\alpha_t(j) = \left[\sum_i \alpha_{t-1}(i) a_{ij} \right] b_j(O_t)$$

$\alpha_1(i)$ (walk):

For Rainy:

$$\alpha_1(R) = 0.6 \times 0.1$$
$$= 0.06$$

For sunny:

$$\alpha_1(S) = 0.4 \times 0.6$$
$$= 0.24$$

$\alpha_2(i)$ (shop):

For Rainy:

$$\alpha_2(R) = [(0.06)(0.7) + (0.24)(0.4)](0.4)$$
$$= (0.042 + 0.096)(0.4)$$
$$= 0.0552$$

For sunny:

$$\alpha_2(S) = [(0.06)(0.3) + (0.24)(0.6)](0.3)$$
$$= (0.018 + 0.144)(0.3)$$
$$= 0.0486$$

$\alpha_3(i)$ (clean):

For Rainy:

$$\alpha_3(R) = [(0.0552)(0.7) + (0.0486)(0.4)](0.5)$$
$$= (0.03864 + 0.01944)(0.5) = 0.02904$$

For sunny:

$$\alpha_3(S) = [(0.0552)(0.3) + (0.0486)(0.6)](0.1)$$

$$= (0.01656 + 0.02916)(0.1)$$

$$= 0.004572$$

sequence likelihood

$$P(O/\lambda) = \alpha_3(R) + \alpha_3(S)$$

$$= 0.02904 + 0.004572$$

$$P(O|\lambda) = 0.033612$$

complexity advantage:

The forward algorithm uses dynamic programming and has complexity $O(TN^2)$ whereas checking all possible state sequences requires $O(N^T)$.

Here, T is the number of observations and N is the number of hidden states.

Task-8:

The viterbi algorithm finds the most probable sequence of hidden states for the given observations.

$$\delta_1(i) = \pi_i b_i(O_1)$$

$$\delta_t(j) = \max_i [\delta_{t-1}(i)\, a_{ij}]\, b_j(O_t)$$

Day 1:

For Rainy:

$$\delta_1(R) = 0.6 \times 0.1 = 0.06$$

For sunny:

$$\delta_1(S) = 0.4 \times 0.6 = 0.24$$

Day 2

For Rainy:

$$\delta_2(R) = \max[(0.06)(0.7), (0.24)(0.4)](0.4)$$

$$= \max[0.042, 0.096] \times 0.4$$

$$= 0.0384$$

Back-pointer: sunny

For sunny:

$$\delta_2(S) = \max[(0.06)(0.3), (0.24)(0.6)](0.3)$$

$$= \max[0.018, 0.144] \times 0.3$$

$$= 0.0432$$

Back-pointer: sunny

Day 3:

For Rainy:

$$\delta_3(R) = \max[(0.0384)(0.7), (0.0432)(0.4)](0.5)$$

$$= \max[0.02688, 0.01728] \times 0.5$$

$$= 0.01344$$

Back-pointer: Rainy

For sunny:

$$\delta_3(S) = \max[(0.0384)(0.3), (0.0432)(0.6)](0.1)$$

$$= \max[0.01152, 0.02592] \times 0.1$$

$$= 0.002592$$

Back-pointer: sunny

Backtracking

The highest final probability is

$$\max(0.01344, 0.002592) = 0.0344$$

The best final state is Rainy

Backtracking gives:

Day 3 = Rainy

Day 2 = Rainy

Day 1 = Sunny

Sunny → Rainy → Rainy

Best path probability:

0.01344

Difference between viterbi and individually most likely states:

viterbi finds the single most probable complete state sequence by considering transitions between states.

choosing the most likely state independently at each time may produce a different sequence and does not necessarily give the most probable complete path

Task-9:

$$\beta_T(i) = 1$$

$$\beta_t(i) = \sum_j a_{ij} b_j(O_{t+1}) \beta_{t+1}(j)$$

$\beta_2(i)$:

$\beta_3(R) = 1$, $\beta_3(S) = 1$

$\beta_2(i)$:

For Rainy:

$$\beta_2(R) = (0.7)(0.5)(1) + (0.3)(0.1)(1)$$

$$= 0.35 + 0.03$$

$$\beta_2(R) = 0.38$$

For sunny:

$$\beta_2(S) = (0.4)(0.5)(1) + (0.6)(0.1)(1)$$

$$= 0.20 + 0.06$$

$$\beta_2(S) = 0.26$$

Posterior probability $\gamma_2(i)$

$$\gamma_t(i) = \frac{\alpha_t(i)\beta_t(i)}{P(O|\lambda)}$$

$\alpha_2(R) = 0.0552$, $\alpha_2(S) = 0.0486$, $P(O|\lambda) = 0.033612$

For Rainy:

$$\gamma_2(R) = \frac{0.0552 \times 0.38}{0.033612}$$

$$= \frac{0.020976}{0.033612}$$

$$\gamma_2(R) = 0.62406$$

For sunny:

$$\gamma_2(S) = \frac{0.0486 \times 0.26}{0.033612}$$

$$= \frac{0.012636}{0.033612}$$

$$\gamma_2(S) = 0.37594$$

Therefore, on Day 2 the weather is more likely to be:

Rainy (62.406%)

Task-10:

Given,

Two token with tags:

$$\{B, I\}$$

unary scores:

Token	B	I
Token 1	2.0	0.5
Token 2	0.5	1.5

Transition scores:

Transition	score
B → B	0.2
B → I	1.0
I → B	0.3
I → I	0.8

(a) scores of all four sequences

$$\text{score} = s_1(y_1) + s_2(y_2) + w(y_1, y_2)$$

1. Sequence BB

$$= 2.0 + 0.5 + 0.2$$

$$\text{score}(BB) = 2.7$$

2. Sequence BI

$$= 2.0 + 1.5 + 1.0$$

$$\text{score}(BI) = 4.5$$

3. Sequence IB

$$= 0.5 + 0.5 + 0.3$$

$$\text{score}(IB) = 1.3$$

4. Sequence II

$$= 0.5 + 1.5 + 0.8$$

$$\text{score}(II) = 2.8$$

(b) Partition function $z(x)$

$$z(x) = \sum_{y} e^{\text{score}(y)}$$

$$= e^{2.7} + e^{4.5} + e^{1.3} + e^{2.8}$$

$$= 14.8797 + 90.0171 + 3.6693 + 16.4446$$

$$z(x) = 125.0108$$

Probability of each sequence:

$$P(y|x) = \frac{e^{\text{score}(y)}}{z(x)}$$

BB:

$$P(BB) = \frac{e^{2.7}}{125.0108}$$

$$= 0.1190$$

BI:

$$P(BI) = \frac{e^{4.5}}{125.0108}$$

$$= 0.7201$$

IB:

$$P(IB) = \frac{e^{1.3}}{125.0108}$$

$$= 0.0994$$

II:

$$P(II) = \frac{e^{2.8}}{125.0108}$$

$$= 0.1315$$

(c) Best sequence

The highest probability is

$$P(BI) = 0.7201$$

∴ Best sequence = BI

why does CRF avoid label-bias and independence assumptions of an HMM?

→ CRF uses global normalization over complete label sequences

→ It can consider dependencies between neighbouring labels.

→ It can use multiple overlapping features of the input

→ It does not require the generative emission and transition independence assumptions of an HMM.

→ Global normalization helps avoid the label-bias problem associated with locally normalized sequence models.

Task-II:

Given,

Number of records:

$$N = 200$$

M_1:

$K = 7$ parameters

Log-likelihood $= -420$

M_2:

$K = 15$ parameters

Log-likelihood $= -410$

BIC:

$$BIC = \ln L - \frac{K}{2} \ln N$$

$$\ln(200) = 5.2983$$

For M_1:

$$BIC_1 = -420 - \frac{7}{2}(5.2983)$$

$$= -420 - 18.5441$$

$$BIC_1 = -438.5441$$

For M_2:

$$BIC_2 = -410 - \frac{15}{2}(5.2983)$$

$$= -410 - 39.7374$$

$$BIC_2 = -449.7374$$

BIC selects highest score

$\therefore$ BIC selects M_1

AIC:

$$AIC = \ln L - k$$

For M1:

$$AIC_1 = -420 - 4$$

$$AIC_1 = -427$$

For M2:

$$AIC_2 = -410 - 15$$

$$AIC_2 = -425$$

AIC selects highest score

∴ AIC selects M_2

Final choice and justification:

→ BIC selects M_1 because it gives a stronger penalty for model complexity.

→ AIC selects M_2 because its higher likelihood compensates for the additional parameters under AIC penalty.

→ M_1 is simpler and has a lower risk of overfitting.

→ M_2 may provide better predictive performance if the additional parameters capture useful patterns

→ M_1 is a reasonable conservative choice for generalisation because BIC favours simpler model

→ However, AIC favours M_2, so validation on unseen data would help make the final decision

Task-12:

Given variables:

Pollution(P), smoke(S), cancer(C), X-ray(X), Dyspnoea(D)

Probabilities:

$P(P=1) = 0.1$; $P(S=1) = 0.3$

cancer probabilities:

P	S	P(C=1 | P,S)
1	1	0.05
1	0	0.02
0	1	0.03
0	0	0.001

X-ray probabilities:

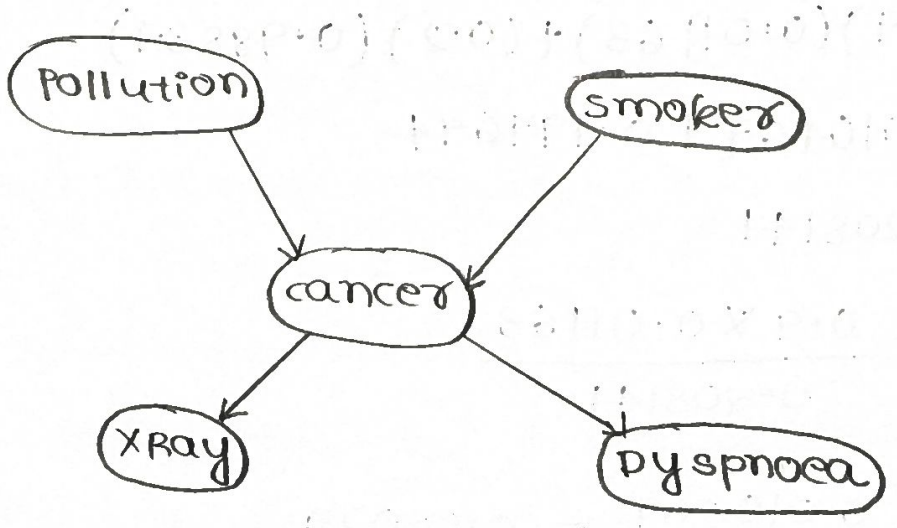
$P(X=1 \mid C=1) = 0.9$; $P(X=1 \mid C=0) = 0.2$

Dyspnoea probabilities:

$P(D=1 \mid C=1) = 0.65$; $P(D=1 \mid C=0) = 0.30$

(a) Network structure and factorised joint distribution

Network structure is:



edges:

$P \to C$; $S \to C$; $C \to X$; $C \to D$

Factorised joint distribution:

$$P(P,S,C,X,D)$$

$$= P(P)\,P(S)\,P(C|P,S)\,P(X|C)\,P(D|C)$$

(b) prior probability $P(C=1)$

using marginalisation:

$$P(C=1) = \sum_{P,S} P(P)\,P(S)\,P(C|P,S)$$

$$= (0.1)(0.3)(0.05) + (0.1)(0.7)(0.02) + (0.9)(0.3)(0.03) + (0.9)(0.7)(0.001)$$

$$= 0.0015 + 0.0014 + 0.0081 + 0.00063$$

$$P(C=1) = 0.01163$$

(c) positive x-ray is observed. compute $P(C=1|X=1)$

using Baye's Theorem:

$$P(C=1|X=1) = \frac{P(X=1|C=1)\,P(C=1)}{P(X=1)}$$

$$P(X=1) = P(X=1|C=1)\,P(C=1) + P(X=1|C=0)\,P(C=0)$$

$$= (0.9)(0.01163) + (0.2)(0.98837)$$

$$= 0.010467 + 0.197674$$

$$= 0.208141$$

$$P(C=1|X=1) = \frac{0.9 \times 0.01163}{0.208141}$$

$$= \frac{0.010467}{0.208141} = 0.05029$$

∴ Probability of cancer after positive x-ray is approx 5.03%

(d) compute $P(C=1 | X=1, D=1)$

using Baye's theorem:

$$P(C|X,D) = \frac{P(X,D|C)P(C)}{P(X,D)}$$

since X and D are conditionally independent given C:

$$P(X,D|C) = P(X|C)\,P(D|C)$$

$$P(C=1|X=1,D=1) = \frac{P(C)(0.9)(0.65)}{P(C)(0.9)(0.65) + P(C=0)(0.2)(0.30)}$$

$$= \frac{(0.01163)(0.9)(0.65)}{(0.01163)(0.9)(0.65) + (0.98837)(0.2)(0.30)}$$

$$= \frac{0.00680355}{0.00680355 + 0.0593022}$$

$$= \frac{0.00680355}{0.06610575}$$

$$P(C=1|X=1,D=1) = 0.10292$$

∴ The probability of cancer is approx 10.29%.

(e) Explain conditional independence and calculate $P(C=1|X=1,S=1)$
why are X and D conditionally independent given C?

-from Bayesian network

$$X \leftarrow C \rightarrow D$$

Cancer is the common parent of X-ray and dyspnoea.
once cancer is known, the x-ray result does not provide additional info about dyspnoea, and vice versa.

$$\therefore X \perp D \mid C$$

calculate $P(C=1|X=1,S=1)$

$$P(C=1|S=1) = P(P=1)\,P(C=1|P=1,S=1) + P(P=0)\,P(C=1|P=0,S=1)$$

$$= (0.1)(0.05) + (0.9)(0.03)$$

$$= 0.005 + 0.027$$

$$= 0.032$$

$$P(C=1|X=1,S=1) = \frac{P(X=1|C=1)\,P(C=1|S=1)}{P(X=1|S=1)}$$

$$= \frac{(0.9)(0.032)}{(0.9)(0.032) + (0.2)(0.968)}$$

$$= \frac{0.0288}{0.0288 + 0.1936}$$

$$= \frac{0.0288}{0.2224}$$

$$P(C=1|X=1,S=1) = 0.12950$$

∴ probability of cancer for a smoker with a positive x-ray is approximately 12.95%.